



Name:

2015

TRIAL
HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

- **General Instructions**
- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I

Pages 3 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 7 – 13

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

1. Evaluate $\int \frac{dx}{x^2 - 4x + 13}$

(A) $\frac{1}{3} \tan^{-1} \left(\frac{x-2}{3} \right) + c$

(B) $\frac{2}{3} \tan^{-1} \left(\frac{x-2}{3} \right) + c$

(C) $\frac{1}{3} \tan^{-1} \left(\frac{2x-4}{3} \right) + c$

(D) $\frac{2}{3} \tan^{-1} \left(\frac{2x-4}{3} \right) + c$

2. Find the equations of the directrices of the ellipse $\frac{x^2}{4} + y^2 = 1$

(A) $x = \pm \frac{2}{\sqrt{5}}$

(B) $x = \pm \frac{4}{\sqrt{3}}$

(C) $x = \pm \sqrt{3}$

(D) $x = \pm \frac{\sqrt{5}}{2}$

3. What is the gradient of the curve $xy - x^2 + 3 = 0$ at the point when $x = 1$?
- (A) -4
- (B) -1
- (C) 1
- (D) 4
4. The region bounded by the curves $y = x^2$ and $y = x^3$ in the first quadrant is rotated about the y -axis. Which integral could be used to find the volume of the solid of revolution formed?
- (A) $V = \pi \int_0^1 \left(y^{\frac{1}{3}} - y^{\frac{1}{2}} \right) dy$
- (B) $V = \pi \int_0^1 \left(y^{\frac{1}{2}} - y^{\frac{1}{3}} \right) dy$
- (C) $V = \pi \int_0^1 \left(y^{\frac{2}{3}} - y \right) dy$
- (D) $V = \pi \int_0^1 (x^4 - x^6) dx$
5. What are the five fifth roots of $1 + \sqrt{3}i$?
- (A) $2^{\frac{1}{5}} \operatorname{cis} \left(\frac{2k\pi}{5} + \frac{\pi}{30} \right), k = 0, 1, 2, 3, 4$
- (B) $2^5 \operatorname{cis} \left(\frac{2k\pi}{5} + \frac{\pi}{30} \right), k = 0, 1, 2, 3, 4$
- (C) $2^{\frac{1}{5}} \operatorname{cis} \left(\frac{2k\pi}{5} + \frac{\pi}{15} \right), k = 0, 1, 2, 3, 4$
- (D) $2^5 \operatorname{cis} \left(\frac{2k\pi}{5} + \frac{\pi}{15} \right), k = 0, 1, 2, 3, 4$

6.

Find $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^5 x \, dx$

(A) 0

(B) 2

(C) $\frac{\pi}{8}$

(D) $\frac{3\pi}{8}$

7. A, B and C are three consecutive terms in an arithmetic progression.

Which of the following is a simplification of $\frac{\sin(A+C)}{\sin B}$?

(A) $2 \cos B$

(B) $\sin 2B$

(C) $\cot B$

(D) 1

8. Consider the graph of the function $x^3 - y^3 = 1$.
- Which of the following statements is NOT true?
- (A) The graph has a vertical tangent at $x = 1$
 - (B) The graph has a horizontal tangent at $x = 0$
 - (C) The line $y = -x$ is an axis of symmetry
 - (D) There is a least one point $P(a, b)$ on the graph such that $b > a$
9. What is the remainder when $P(x) = x^3 + x^2 - x + 1$ is divided by $(x - 1 - i)$?
- (A) $-3i - 2$
 - (B) $3i - 2$
 - (C) $3i + 2$
 - (D) $2 - 3i$
10. Solve the inequality: $\frac{x+1}{x-3} \leq \frac{x+3}{x-2}$.
- (A) $x < 2$ and $x > 3$
 - (B) $x < 2$ and $3 < x \leq 7$
 - (C) $2 < x < 3$
 - (D) $2 < x < 3$ and $x \geq 7$

End of Section I

Section II
90 marks
Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Let $A = 3 + 3\sqrt{3}i$ and $B = -5 - 12i$.

Find the value of:

- | | | |
|-------|---------------------------------|---|
| (i) | $\bar{B}$ | 1 |
| (ii) | $\frac{A}{B}$ | 2 |
| (iii) | The square roots of B | 2 |
| (iv) | The modulus and argument of A | 2 |
| (v) | A^4 | 1 |

(b) The roots of the polynomial equation $2x^3 - 3x^2 + 4x - 5 = 0$ are α , β and γ . Find the polynomial equation which has roots:

- | | | |
|------|---|---|
| (i) | $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$ | 2 |
| (ii) | 2α , 2β and 2γ | 2 |

(c)	Find $\int \frac{dx}{\sqrt{9 + 16x - 4x^2}}$	3
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End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_0^{\frac{\sqrt{\pi}}{2}} 3x \sin(x^2) dx$ **3**

(b) (i) Find the values of A , B , and C such that: **2**

$$\frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+2}$$

(ii) Hence evaluate $\int_2^4 \frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} dx$ **2**

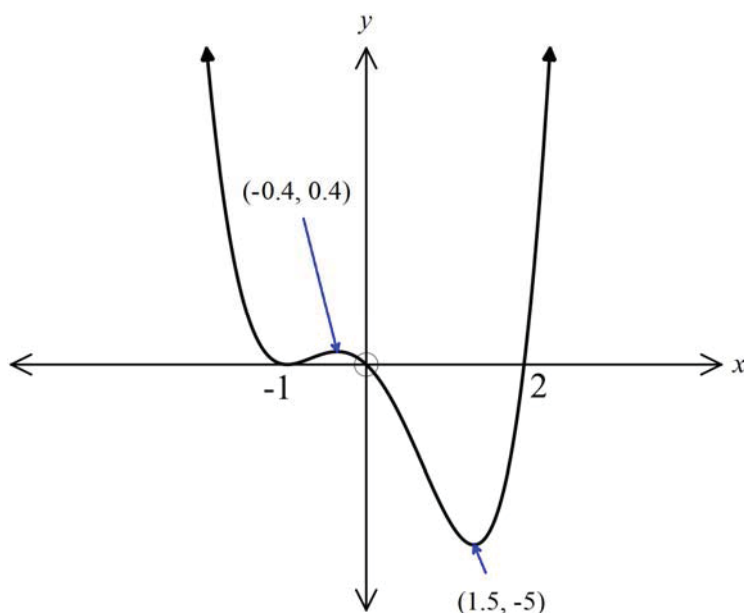
(c) Solve the equation $x^4 - 7x^3 + 17x^2 - x - 26 = 0$, given that $x = (3 - 2i)$ is a root of the equation. **4**

(d) Use the method of cylindrical shells to find the volume of the solid generated by revolving the region enclosed by $y = 3x^2 - x^3$ and the x axis around the y -axis. **4**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) The graph of $y = f(x)$ is shown below.



Draw separate $\frac{1}{3}$ page sketches for each of the following:

(i) $y = |f(x)|$ 1

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y^2 = f(x)$ 2

(iv) $y = e^{f(x)}$ 2

(b) Show that: $\frac{\cos A - \cos(A + 2B)}{2 \sin B} = \sin(A + B)$ 3

(c) A particle is projected vertically upwards. The resistance to the motion is proportional to the square of the velocity. The velocity of projection is V m/s.

(i) Show that the acceleration is given by: $\ddot{x} = -(g + Kv^2)$ where K is a constant. 1

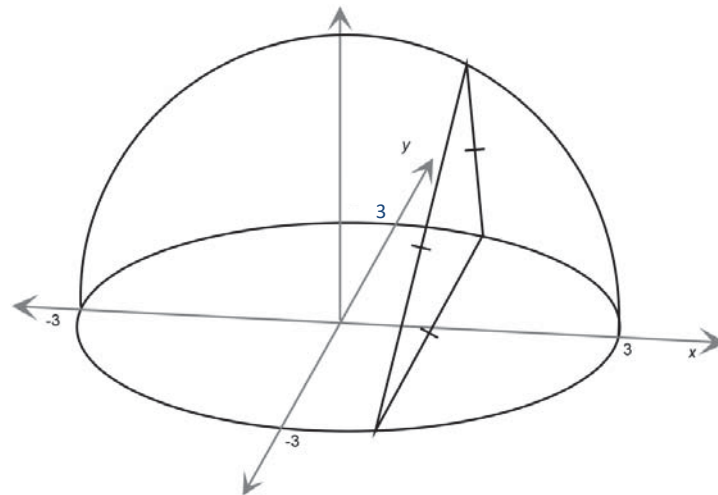
(ii) Find the maximum height reached and the time taken to reach this height, expressing your answer in terms of V and K . 4

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a)

4



The diagram above shows a solid which has the circle $x^2 + y^2 = 9$ as its base.

The cross-section perpendicular to the x axis is an equilateral triangle.

Calculate the volume of the solid.

(b)

Given that $x^4 - 6x^3 + 9x^2 + 4x - 12 = 0$, has a double root at $x = \alpha$,
find the value of α .

3

(c)

A sequence is defined such that $u_1 = 1, u_2 = 1$ and $u_n = u_{n-1} + u_{n-2}$ for $n \geq 3$.

4

Prove by induction that $u_n < \left(\frac{7}{4}\right)^n$ for integers $n \geq 1$.

(d)

The point $P\left(ct, \frac{c}{t}\right)$ lies on the rectangular hyperbola $xy = c^2$.

(i) Show that the normal at P cuts the hyperbola again at Q with coordinates

3

$$\left(-\frac{c}{t^3}, -ct^3\right)$$

(ii) Hence find the coordinates of the point R where the normal at Q cuts the hyperbola again.

1

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) z represents the complex number $x + iy$. Sketch the regions:

(i) $|\arg z| < \frac{\pi}{4}$ 1

(ii) $\operatorname{Im}(z^2) = 4$ 2

(b) The complex roots of $z^3 = 1$ are 1 , ω and ω^2 .

(i) Find the value of $(1 + \omega^2)^6$ 1

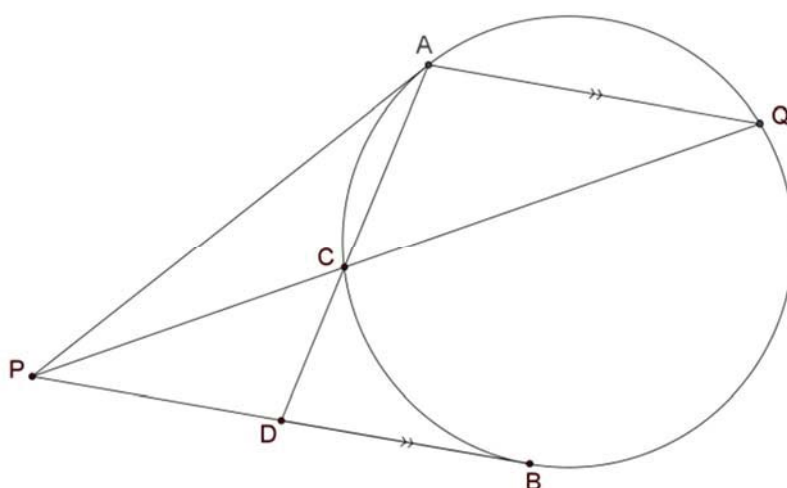
(ii) Hence show that $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^5) = 9$ 2

(c) In the diagram below, PA and PB are tangents to the circle. The chord AQ is parallel to the tangent PB . PCQ is a secant to the circle and chord AC produced meets PB at D .

(i) Show that $\triangle CDP$ is similar to $\triangle PDA$. 2

(ii) Hence show that $PD^2 = AD \times CD$. 1

(iii) Hence, or otherwise, prove that AD bisects PB . 2



Question 15 continues on the page 11

Question 15 (continued)

(d) Derive the reduction formula:

4

$$\int x^n e^{-x^2} dx = -\frac{1}{2} x^{n-1} e^{-x^2} + \frac{n-1}{2} \int x^{n-2} e^{-x^2} dx$$

and use this reduction formula to evaluate $\int_0^1 x^5 e^{-x^2} dx$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) Consider the hyperbola with equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. P is the point $(a \sec \theta, b \tan \theta)$.

(i) Show that the equation of the tangent at P is $bx \sec \theta - ay \tan \theta = ab$. 2

(ii) Find the equation of the normal at P . 2

(iii) Find the coordinates of the points A and B where the tangent and normal respectively cut the y -axis. 2

(iv) Show that AB is the diameter of the circle that passes through the foci of the hyperbola. 3

(b) Suppose n is a positive integer.

(i) Show that $1 - x^2 + x^4 - x^6 + \dots + (-1)^{n-1} x^{2n-2} = \frac{1 - (-1)^n x^{2n}}{1 + x^2}$ 1

(ii) Hence show that $-x^{2n} \leq \frac{1}{1+x^2} - \left(1 - x^2 + x^4 - x^6 + \dots + (-1)^{n-1} x^{2n-2}\right) \leq x^{2n}$ 2

(iii) By integrating over suitable values of x , deduce that 2

$$-\frac{1}{2n+1} \leq \frac{\pi}{4} - \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots + (-1)^{n-1} \frac{1}{2n-1}\right) \leq \frac{1}{2n+1}$$

(iv) Explain why $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$ 1

End of Examination.

MATHS EXT. 2: TRIAL 2015

SECTION I:

① A

$$\begin{aligned} \int \frac{dx}{x^2 - 4x + 13} \\ &= \int \frac{dx}{(x-2)^2 + 9} \\ &= \frac{1}{3} \tan^{-1} \left(\frac{x-2}{3} \right) + C \end{aligned}$$

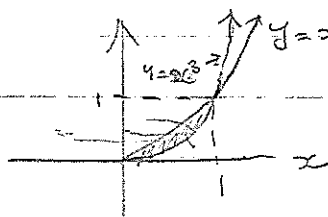
② B

$$\begin{aligned} \frac{x^2}{4} + y^2 &= 1 \\ a &= 2, \quad b = 1 \\ b^2 &= a^2(1 - e^2) \\ 1 - e^2 &= \frac{1}{4} \Rightarrow e^2 = \frac{3}{4} \\ e &= \frac{\sqrt{3}}{2} \\ x &= \pm \frac{a}{e} = \pm \frac{2}{\frac{\sqrt{3}}{2}} \\ &= \pm \frac{4}{\sqrt{3}} \end{aligned}$$

③ D

$$\begin{aligned} xy - x^2 + 3 &= 0 \\ 2x \frac{dy}{dx} + y - 2x &= 0 \\ \frac{dy}{dx} &= \frac{2x - y}{2x} \\ x=1, &\Rightarrow y = -2 \\ \therefore m &= \frac{2(1) + 2}{1} = 4 \end{aligned}$$

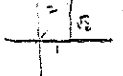
④ C



$$\begin{aligned} SV &= \pi (x_2^2 - x_1^2) \delta y \\ V &= \pi \int_0^1 (y^{\frac{1}{3}})^2 - (y^{\frac{2}{3}})^2 dy \\ &= \pi \int_0^1 y^{\frac{2}{3}} - y dy \end{aligned}$$

⑤ C

$$1 + \sqrt{3}i = 2 \operatorname{cis} \frac{\pi}{3}$$



$$\text{Let } z = r \operatorname{cis} \theta$$

$$z^5 = 2 \operatorname{cis} \frac{\pi}{3}$$

$$r^5 \operatorname{cis} 5\theta = 2 \operatorname{cis} \frac{\pi}{3}$$

$$r = 2^{\frac{1}{5}}$$

$$\text{and } 5\theta = \frac{\pi}{3} + 2k\pi$$

$$\theta = \frac{\pi}{15} + \frac{2k\pi}{5}$$

⑥ A

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^5 x \, dx \\ \downarrow \\ \text{odd function} \\ \therefore \int_{-a}^a = 0 \end{aligned}$$

⑦

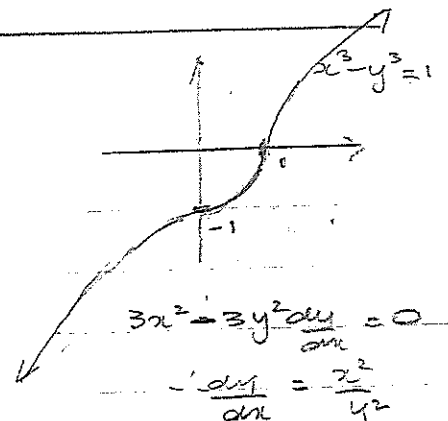
A

$$\frac{A+C}{2} = 2$$

$$\therefore A+C = 2B$$

$$\begin{aligned} \frac{\sin 2A}{\sin B} &= \frac{\sin 2C}{\sin B} \\ &= 2 \sin B \end{aligned}$$

⑧ D



A: $x=1, y=0 \Rightarrow$ vertical tangent V.

B: $x=0, y' = 0 \Rightarrow$ horizontal tangent H.

C: Sum of the absolute values $y = -x$ ✓

D: $P(a, b), b > a \Rightarrow y > x$

$$\begin{aligned} y^3 &= x^3 - 1 \quad \therefore y^3 < x^3 \\ &\Rightarrow y < x \end{aligned}$$

X

(9) B

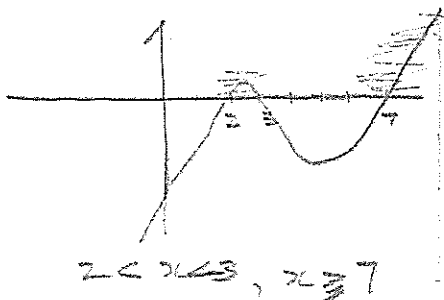
$$\begin{aligned} P(1+i) &= (1+i)^3 + (1+i)^2 - (1+i) + 1 \\ &= 1+3i-3-i+1+2i-1-1-i+1 \\ &= -2+3i \end{aligned}$$

(10) D

$$\frac{x+1}{x-3} \leq \frac{x+3}{x-2}$$

$$\begin{aligned} (x+1)(x-3)(x-2)^2 &\leq (x+3)(x-2)(x-3)^2 \\ (x+3)(x-2)(x-3)^2 - (x+1)(x-3)(x-2)^2 &\geq 0 \\ (x-2)(x-3) \left[(x+3)(x-3) - (x+1)(x-2) \right] &\geq 0 \end{aligned}$$

$$\begin{aligned} (x-2)(x-3) [x^2-9-x^2+x+2] &\geq 0 \\ (x-2)(x-3)(x-7) &\geq 0 \end{aligned}$$



$$2 < x < 3, x \geq 7$$

(a) cont

(ii) Let $(x+y)^2 = -5-12i$

$$x^2 + y^2 = -5 \quad \text{--- (1)}$$

$$2xy = -12$$

$$xy = -6 \quad \text{--- (2)}$$

Sub $y = -\frac{6}{x}$ in (1)

$$x^2 - \left(-\frac{6}{x}\right)^2 = -5$$

$$x^4 - 36 = -5x^2$$

$$x^4 + 5x^2 - 36 = 0$$

$$(x^2+9)(x^2-4) = 0$$

$$\therefore x = \pm 2 \quad (x \in \mathbb{R})$$

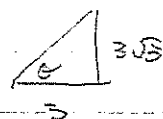
$$x=2, y=-3 \text{ or } x=-2, y=3$$

$\therefore$ Roots are $2-3i$ or $-2+3i$
 $[\pm(2-3i)]$

(iv) $r \cos \theta = 3+3\sqrt{3}i$

$$r = 3(2) = 6$$

$$\theta = \frac{\pi}{3}$$



$$\therefore 3+3\sqrt{3}i = 6 \cos \frac{\pi}{3}$$

$$[\text{mod } A = 2, \text{arg } A = \frac{\pi}{3}]$$

(v) $A^4 = 6^4 \cos \frac{4\pi}{3}$

$$= -1296 \cos \frac{\pi}{3}$$

$$= -1296 \left[\frac{1}{2} + i \frac{\sqrt{3}}{2} \right]$$

$$= -648 [1 + i\sqrt{3}]$$

SECTION II

SECTION II

(a) (i) $\bar{B} = -5+12i$

$$(ii) \frac{A}{B} = \frac{3+3\sqrt{3}i}{-5-12i} \times \frac{-5+12i}{-5+12i}$$

$$= \frac{-15+36i-15\sqrt{3}i-36\sqrt{3}}{25+144}$$

$$= \frac{(-15-36\sqrt{3}) + (36-15\sqrt{3})i}{169}$$

QUESTION 11 (cont)

(b) $2x^3 - 3x^2 + 4x - 5 = 0$

(c) Equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$

is $2\left(\frac{1}{x}\right)^3 - 3\left(\frac{1}{x}\right)^2 + 4\left(\frac{1}{x}\right) - 5 = 0$

$$2 - 3x + 4x^2 - 5x^3 = 0$$

$$5x^3 - 4x^2 + 3x - 2 = 0$$

(11) Equation with roots $2\alpha, 2\beta, 2\gamma$ is

$$2\left(\frac{x}{2}\right)^3 - 3\left(\frac{x}{2}\right)^2 + 4\left(\frac{x}{2}\right) - 5 = 0$$

$$\frac{2x^3}{8} - \frac{3x^2}{4} + \frac{4x}{2} - 5 = 0$$

$$x^3 - 3x^2 + 8x - 20 = 0$$

$$\begin{aligned} (c) \quad & \int \frac{dx}{\sqrt{9 + 16x - 4x^2}} \\ &= \int \frac{dx}{\sqrt{9 - [4x^2 - 16x]}} \\ &= \int \frac{-dx}{\sqrt{9 - 4[x - 2]^2 + 16}} \\ &= \int \frac{dx}{2\sqrt{\frac{25}{4} - (x-2)^2}} \\ &= \frac{1}{2} \sin^{-1} \frac{2}{5} (x-2) + C \end{aligned}$$

QUESTION 12

$$\begin{aligned} (a) \quad & \int_0^{\frac{\sqrt{\pi}}{2}} 3x \sin(x^2) dx \\ &= \int_0^{\frac{\sqrt{\pi}}{2}} \sin(x^2) 2x dx \quad u = x^2 \\ & \quad \quad \quad du = 2x dx \\ &= \int_0^{\frac{\pi}{4}} \sin u du \\ &= \left[-\cos u \right]_0^{\frac{\pi}{4}} \\ &= \left[-\frac{1}{\sqrt{2}} + 1 \right] = \frac{\sqrt{2}-1}{\sqrt{2}} \end{aligned}$$

(b) $\frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+2}$

$$\begin{aligned} A(x-1)(x+2) + Bx(x+2) + Cx(x-1) \\ = 4x^2 - 3x - 4 \end{aligned}$$

$$\begin{aligned} (A+B+C)x^2 + (A+2B-C)x - 2A \\ = 4x^2 - 3x - 4 \end{aligned}$$

$$A+B+C = 4 \quad \text{--- (1)}$$

$$A+2B-C = -3 \quad \text{--- (2)}$$

$$-2A = -4 \Rightarrow A = 2$$

$$\text{In (1)} \quad B+C = 2 \quad \text{--- (1A)}$$

$$\text{In (2)} \quad 2B-C = -5 \quad \text{--- (2A)}$$

$$(1A) + (2A) \Rightarrow 3B = -3$$

$$B = -1$$

$$\therefore C = 3$$

$$\therefore \boxed{A=2, B=-1, C=3}$$

QUESTION 12: (continued)

$$\begin{aligned}
 (b) \quad (ii) \quad \int_2^4 \frac{4x^2 - 3x - 1}{x^3 + x^2 - 2x} dx &= \int_2^4 \left(\frac{7}{x} - \frac{1}{x-1} + \frac{3}{x+2} \right) dx \\
 &= \left[7 \ln x - \ln(x-1) + 3 \ln(x+2) \right]_2^4 \\
 &= [2 \ln 4 + \ln 3 + 3 \ln 6 - 2 \ln 2 - \ln 1 - 3 \ln 4] \\
 &= \ln \left[\frac{4^2 \times 3 \times 6^3}{2^2 \times 4^2} \right] = \ln \left[\frac{81}{2} \right]
 \end{aligned}$$

$$(c) \quad x^4 - 7x^3 + 17x^2 - x - 26 = 0$$

real coeff. $x = 3 - 2i$ is a root

$$2x = 6$$

$\therefore x = 3 + 2i$ is a root (1)

$$4x = 13$$

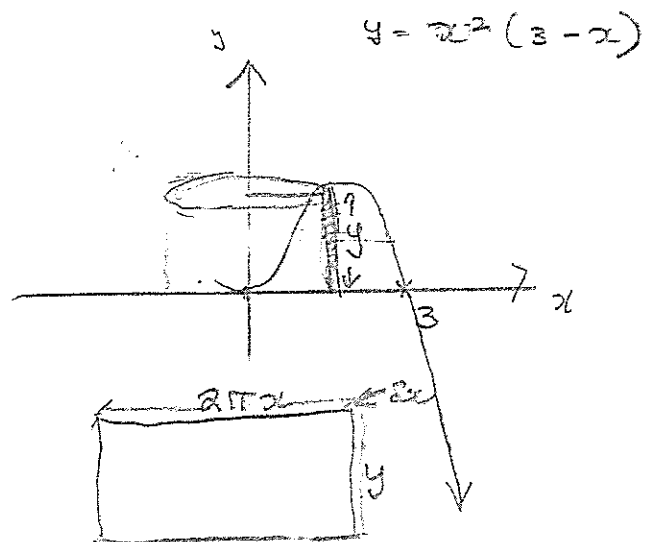
$\therefore x^2 - 6x + 13 = 0$ is a factor

By observation (or long division)

$$\begin{aligned}
 x^4 - 7x^3 + 17x^2 - x - 26 &= (x^2 - 6x + 13)(x^2 - x - 2) \\
 &= (x^2 - 6x + 13)(x - 2)(x + 1)
 \end{aligned}$$

$\therefore$ Roots are $x = -2, 2$ and $3 \pm 2i$

$$\begin{aligned}
 (d) \quad V &= \int_0^3 2\pi xy \, dx \\
 &= \int_0^3 2\pi (3x^2 - x^4) \, dx \\
 &= 2\pi \left[\frac{3}{4} x^4 - \frac{x^5}{5} \right]_0^3 \\
 &= 2\pi \left[\frac{3}{4} \times 81 - \frac{3 \times 81}{5} \right] \\
 &= 2\pi \times \frac{3}{20} \times 81 = \frac{243}{10} \pi \, u^3
 \end{aligned}$$

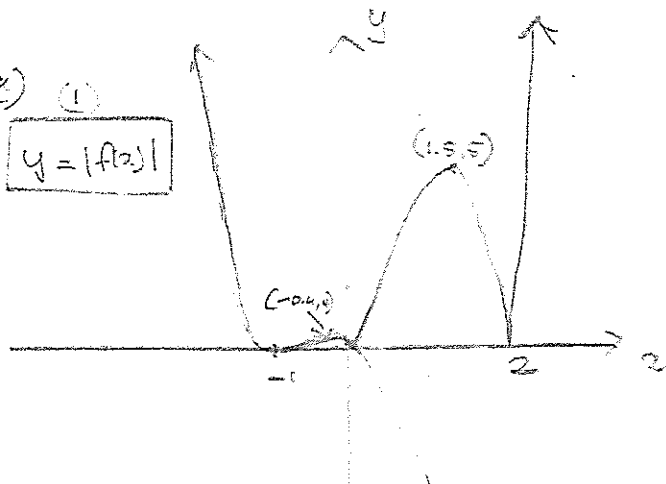


$$\delta V = 2\pi xy \delta x$$

QUESTION 13:

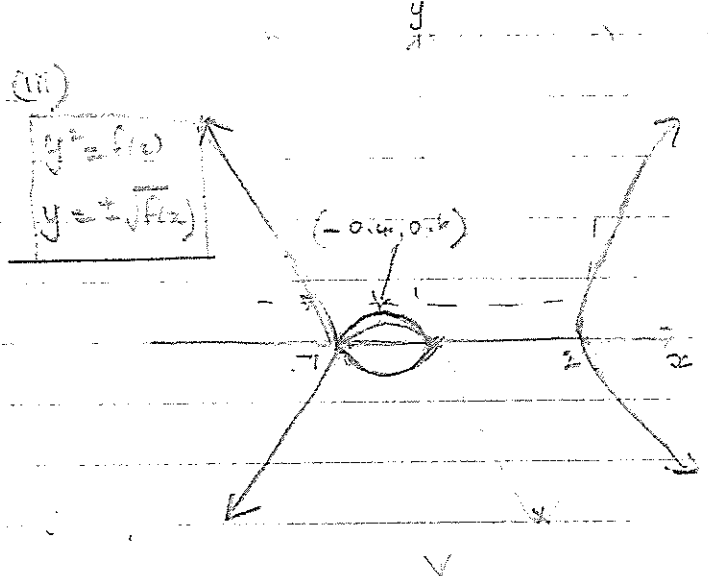
(c) (i)

$$y = |f(x)|$$



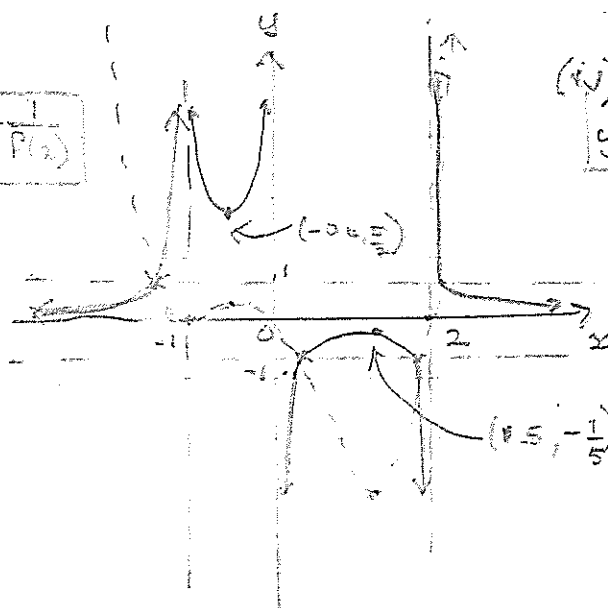
(ii)

$$y^2 = f(x) \\ y = \pm \sqrt{f(x)}$$



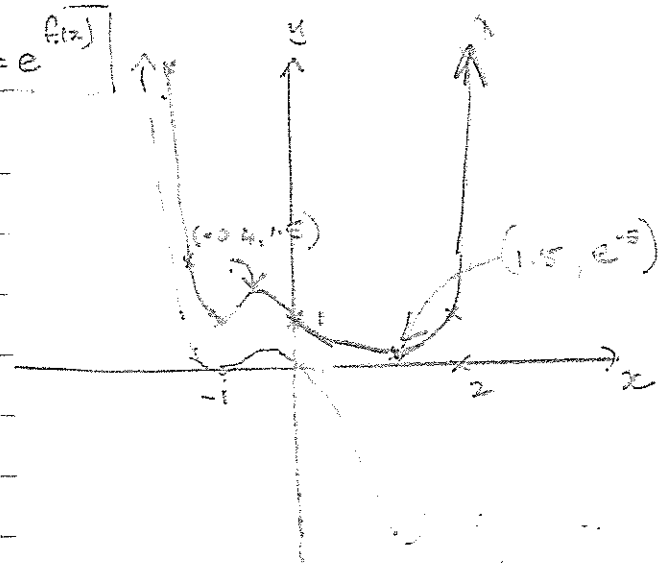
(iii)

$$y = \frac{1}{f(x)}$$



(iv)

$$y = e^{f(x)}$$



$$(b) \quad LHS = \frac{\cos A - [\cos A \cos 2B - \sin A \sin 2B]}{2 \sin B}$$

$$= \frac{\cos A - [\cos A (1 - 2 \sin^2 B) - \sin A \cdot 2 \cos B \sin B]}{2 \sin B}$$

$$= \frac{\cos A - \cos A + 2 \cos A \sin^2 B + 2 \sin A \cos B \sin B}{2 \sin B}$$

$$= \frac{\cos A \sin B + \sin A \cos B}{\sin B}$$

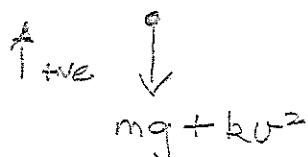
$$= \sin(A+B)$$

QUESTION 13

(c) (i)

$$F = ma$$

$$-(mg + kv^2) = m\ddot{x}$$



$$\ddot{x} = -\left(g + \frac{k}{m}v^2\right)$$

$$= -\left(g + Kv^2\right) \text{ where } K \text{ is a constant}$$

(ii) Need v, x

$$t=0, v=V, x=0.$$

$$v \frac{dv}{dx} = -(g + Kv^2)$$

$$\frac{dv}{dx} = -\frac{(g + Kv^2)}{v}$$

$$\therefore \frac{dx}{dv} = -\frac{v}{g + Kv^2}$$

$$\begin{aligned} x &= -\frac{1}{2K} \int \frac{2Kv}{g + Kv^2} dv \\ &= -\frac{1}{2K} \ln(g + Kv^2) + C \end{aligned}$$

$$x=0, v=V \Rightarrow C = \frac{1}{2K} \ln(g + KV^2)$$

$$\therefore x = -\frac{1}{2K} \ln\left(\frac{g + Kv^2}{g + KV^2}\right)$$

$$\text{when } v=0, x_{\max} = -\frac{1}{2K} \ln\left(\frac{g}{g + KV^2}\right)$$

$$= \frac{1}{2K} \ln\left(1 + \frac{KV^2}{g}\right) \text{ m.}$$

$$\text{Need } t, b. \quad \frac{dv}{dt} = -(g + Kv^2)$$

$$t=0, v=V$$

$$\frac{dt}{dV} = -\frac{1}{g + Kv^2}$$

$$\therefore = -\frac{1}{K\left(\frac{g}{K} + v^2\right)}$$

QUESTION 13 (continued)

(5) (ii) $\therefore t = -\frac{1}{K} \sqrt{\frac{K}{g}} \tan^{-1}\left(\sqrt{\frac{K}{g}} v\right) + C$ (1)

$t=0, v=V \Rightarrow C = \frac{1}{\sqrt{Kg}} \tan^{-1} \sqrt{\frac{K}{g}} V$

$\therefore t = \frac{1}{\sqrt{Kg}} \left[\tan^{-1} \sqrt{\frac{K}{g}} V - \tan^{-1} \sqrt{\frac{K}{g}} v \right]$

$v=0 \Rightarrow T = \frac{1}{\sqrt{Kg}} \tan^{-1} \sqrt{\frac{K}{g}} V \text{ secs}$

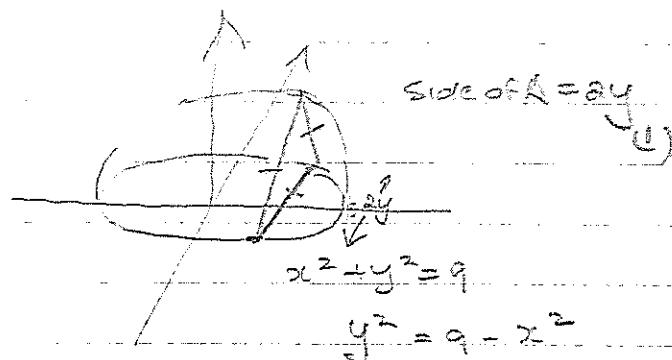
QUESTION 14

(a)

$A_A = \frac{1}{2} (By)(2y) \sin 60^\circ$
 $= 2y^2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3} y^2$ (1)

$V = 2 \int_0^3 \sqrt{3} y^2 dx$

$= 2\sqrt{3} \int_0^3 (9 - x^2) dx$ (1) $= 2\sqrt{3} \left[9x - \frac{x^3}{3} \right]_0^3 = 2\sqrt{3} \left[27 - \frac{27}{3} \right]$
 $= 36\sqrt{3} \text{ u}^3$



(b) $f(x) = x^4 - 6x^3 + 9x^2 + 4x - 12$

$f'(x) = 4x^3 - 18x^2 + 18x + 4$

$f'(x) = 0 \Rightarrow 2x^3 - 9x^2 + 9x + 2 = 0$ $f(1) \neq 0, f(-1) = 0$

$f'(2) = 2(8) - 9(4) + 9(2) + 2 = 0$

$\therefore x=2$ is a root of $f'(x)$

Also $f(2) = 16 - 6(8) + 9(4) + 4(2) - 12 = 0$

$\therefore x=2$ is at least a double root. But there can't be a

triple root as we know there is a double root & there are only 4 roots

$\therefore x=2$ is a double root and $d=2$

QUESTION 14 (continued)

(C) $u_1 = 1, u_2 = 1, u_n = u_{n-1} + u_{n-2} \quad n \geq 3$

To prove: $u_n < \left(\frac{7}{4}\right)^n \quad n \geq 1$

(A) : $n=1, u_1 = 1 < \left(\frac{7}{4}\right)^1 \therefore S(1) \text{ true.}$
 $n=2, u_2 = 1 < \left(\frac{7}{4}\right)^2 \therefore S(2) \text{ true.}$

(B) : Assume $S(k)$ and $S(k-1)$ true
 $\therefore u_k < \left(\frac{7}{4}\right)^k \text{ and } u_{k-1} < \left(\frac{7}{4}\right)^{k-1}$

RTP: $u_{k+1} < \left(\frac{7}{4}\right)^{k+1}$

LHS = $u_{k+1} = u_k + u_{k-1}$
 $< \left(\frac{7}{4}\right)^k + \left(\frac{7}{4}\right)^{k-1}$

Alternatively (or something similar)
 RTP: $\left(\frac{7}{4}\right)^{k+1} - u_{k+1} > 0$

LHS = $\left(\frac{7}{4}\right)^{k+1} - (u_k + u_{k-1})$
 $> \left(\frac{7}{4}\right)^{k+1} - \left(\frac{7}{4}\right)^k - \left(\frac{7}{4}\right)^{k-1}$
 $= \left(\frac{7}{4}\right)^{k-1} \left[\left(\frac{7}{4}\right)^2 - \frac{7}{4} - 1 \right]$
 $= \left(\frac{7}{4}\right)^{k-1} \left(\frac{5}{6}\right) > 0$

$= \left(\frac{7}{4}\right)^{k-1} \left(\frac{7}{4} + 1\right)$
 $= \left(\frac{7}{4}\right)^{k-1} \left(\frac{11}{4}\right)$

$= \left(\frac{7}{4}\right)^{k-1} \left(\frac{44}{16}\right)$

$< \left(\frac{7}{4}\right)^{k-1} \left(\frac{7}{4}\right)^2 \quad \left(\frac{7}{4}\right)^2 = \frac{49}{16}$

$= \left(\frac{7}{4}\right)^{k+1} \therefore S(k+1) \text{ true}$

(C) $\therefore S(1)$ and $S(2)$ are true $\therefore S(3)$ is true and by the process of mathematical induction, $S(n)$ is true for all n .

(d) (1) $xy = c^2, \frac{dy}{dx} = -\frac{c^2}{x^2}$

At $x=ct \quad \text{M.T.} = -\frac{1}{t^2} \therefore mN = t^2$

$\therefore$ Equation of normal is $y - \frac{c}{t} = t^2(x - ct)$

QUESTION 14 (continued)

(a) (i) (cont) $y = \frac{c}{t} = t^2(x - ct) \quad \text{--- (1)}$

Solve with $y = \frac{c^2}{x} \quad \text{--- (2)}$

$$\Rightarrow \frac{c^2}{x} - \frac{c}{t} = t^2(x - ct) \Rightarrow ct^2 - xc = t^3x^2 - ct^4x$$

$$t^3x^2 + (c - ct^4)x - c^2t = 0$$

$$(t^3x^2 + c)(x - ct) = 0$$

You know

$x = ct$ is a solution

$\therefore$ normal cuts again when $x = -\frac{c}{t^3}$ and $y = -\frac{c^2t^3}{0} = -ct^3$

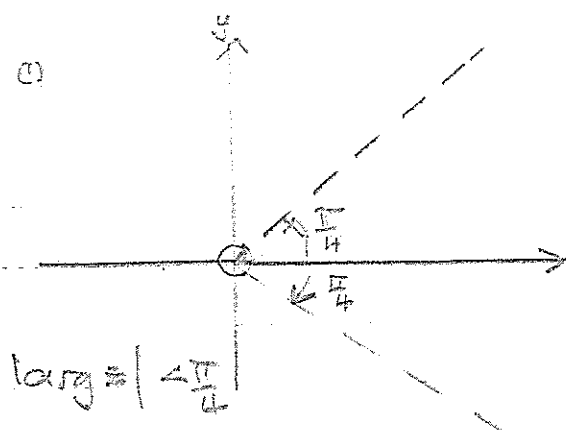
$$\therefore Q = \left[-\frac{c}{t^3}, -ct^3 \right]$$

(i) "t" $= -\frac{1}{t^3}$ at Q. $\therefore$ Normal at Q cuts at R where $R = \left[-\frac{c}{16t^3}, -c(16t)^3 \right]$

$$R = \left[-\frac{c}{\left(-\frac{1}{t^3}\right)^2}, -c\left(-\frac{1}{t^3}\right)^3 \right] = \left[ct^9, \frac{c}{t^9} \right]$$

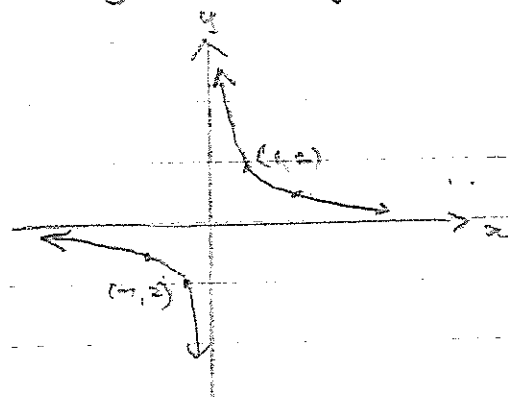
QUESTION 15

(a) (i)



(ii) $\operatorname{Im}(x+iy)^2 = 4$

$\therefore 2xy = 4 \Rightarrow xy = 2$



(b) $\omega^3 = 1$ and $(\omega^2)^3 = 1 \quad z^3 - 1 = 0$

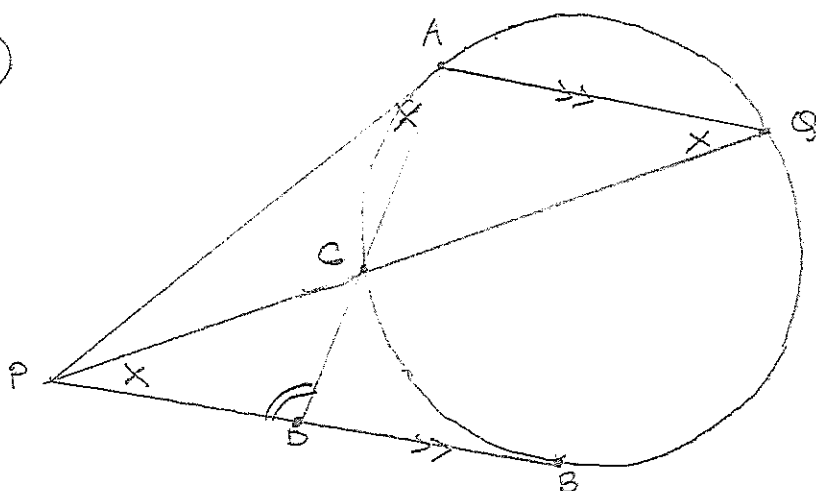
$\therefore$ Sum of roots $= 1 + \omega + \omega^2 = 0 \Rightarrow 1 + \omega^2 = -\omega$

(i) $(1 + \omega^2)^6 = (-\omega)^6 \quad (1) \quad 1 = -\omega - \omega^2$
 $= [(-\omega)^3]^2 = (-1)^2 = 1$

(ii) $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^5) \quad \omega^3 = 1$
 $= (1 - \omega)(1 - \omega^2)(1 - \omega)(1 - \omega^2) \quad (1)$
 $= (1 - \omega)^2(1 - \omega^2)^2 = (1 - \omega - \omega^2 + \omega^3)^2 = (1 + 1 + 1)^2 = 9$

QUESTION 15 (continued)

(c)



(i) In $\triangle PDC$ and ADP ,

$\angle D$ is common

$\angle CPD = \angle CQA$ (alternate $\angle$'s, $AC \parallel PB$)

But $\angle PAO = \angle CQA$ ($\angle$ between chord & tangent equals $\angle$ in alternate segment)

$\therefore \angle CPD = \angle PAO$ (equiangular)

(ii) $\frac{PD}{DA} = \frac{DC}{PD}$ (corresponding sides of similar $\triangle$'s)

$$\therefore PD^2 = AD \cdot DC$$

(iii) But $DB^2 = AD \cdot DC$ (square of tangent equals the product of the intersecting secants)

$$\therefore PD^2 = DB^2$$

and $PD = DB$.

QUESTION 15. (continued)

$$(d) \int x^n e^{-x^2} dx = \int \frac{x^{n+1}}{-2} (-2xe^{-x^2}) dx$$
$$= -\frac{1}{2} \int x^{n+1} (-2xe^{-x^2}) dx$$

$$u = x^{n+1} \quad v' = -2xe^{-x^2}$$
$$u' = (n+1)x^{n-2} \quad v = e^{-x^2}$$

$$= -\frac{1}{2} \left[x^{n+1} e^{-x^2} \right] + \frac{1}{2} \int (n+1) x^{n-2} e^{-x^2} dx$$

$$= -\frac{1}{2} x^{n+1} e^{-x^2} + \frac{(n+1)}{2} \int x^{n-2} e^{-x^2} dx$$

$$\text{Let } I_n = \left[-\frac{1}{2} x^{n+1} e^{-x^2} \right]_0^1 + \frac{n+1}{2} I_{n-2}$$

$$I_5 = \left[-\frac{1}{2} x^6 e^{-x^2} \right]_0^1 + 2 I_3$$

$$= \left(-\frac{1}{2} \cdot e^{-1} \right) + 2 I_3$$

$$I_3 = \left[-\frac{1}{2} x^4 e^{-x^2} \right]_0^1 + 1 \cdot I_1$$

$$= -\frac{1}{2} e^{-1} + I_1$$

$$I_1 = \int_0^1 x e^{-x^2} dx$$

$$= -\frac{1}{2} \left[e^{-x^2} \right]_0^1 = -\frac{1}{2} [e^{-1} - 1]$$

$$\therefore I_3 = -\frac{1}{2} [e^{-1} + e^{-1} - 1]$$

$$I_5 = -\frac{1}{2} [e^{-1}] + 2 \left[-\frac{1}{2} (e^{-1} + e^{-1} - 1) \right]$$

$$= -\frac{1}{2} e^{-1} - (e^{-1} + e^{-1} - 1)$$

$$= -\frac{5}{2} e^{-1} + 1$$

QUESTION 16:

$$(a) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$(i) \quad x = a \sec \theta \quad y = b \tan \theta$$

$$\frac{dx}{d\theta} = a \sec \theta \tan \theta \quad \frac{dy}{d\theta} = b \sec^2 \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= \frac{b \sec^2 \theta}{a \sec \theta \tan \theta} = \frac{b \sec \theta}{a \tan \theta}$$

$\therefore$ Eqn of tangent is

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$ay \tan \theta - ab \tan^2 \theta = b \sec \theta x - ab \sec^2 \theta$$

$$b \sec \theta x - a \tan \theta y = ab (\sec^2 \theta - \tan^2 \theta)$$

$$\therefore b \sec \theta x - a \tan \theta y = ab \quad \text{since } \sec^2 \theta - \tan^2 \theta = 1 \quad (1)$$

$$(ii) \quad \text{Normal at P: } m_N = - \frac{a \tan \theta}{b \sec \theta}$$

$$\therefore \text{Eqn of normal is } y - b \tan \theta = \frac{-a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

$$b \sec \theta y - b^2 \tan \theta \sec \theta = -a \tan \theta x + a^2 \sec \theta \tan \theta$$

$$\therefore a \tan \theta x + b \sec \theta y = (a^2 + b^2) \sec \theta \tan \theta$$

$$\therefore \frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2 \quad (2)$$

$$(iii) \quad \text{Put } x=0 \text{ in } (1) \Rightarrow y = - \frac{ab}{a \tan \theta} \therefore A = \left(0, -\frac{b}{\tan \theta}\right)$$

$$\text{Put } x=0 \text{ in } (2) \Rightarrow y = \frac{(a^2 + b^2) \tan \theta}{b} \therefore B = \left(0, \frac{(a^2 + b^2) \tan \theta}{b}\right)$$

QUESTION 16 (continued)

(a) (iv) $A = (0, -\frac{b}{\tan \theta})$

$B = (0, \frac{(a^2+b^2)\tan \theta}{b})$

Foci $S(ae, 0)$ $S'(-ae, 0)$

Need to show: $m_{AS} \times m_{BS} = -1$

$$m_{AS} = \frac{\frac{b}{\tan \theta}}{ae} = \frac{b}{ae \tan \theta}$$

$$m_{BS} = \frac{\frac{(a^2+b^2)\tan \theta}{b}}{-ae} = \frac{(a^2+b^2)\tan \theta}{-abe}$$

$$m_{AS} \times m_{BS} = \frac{b}{ae \tan \theta} \times -\frac{(a^2+b^2)\tan \theta}{abe}$$

$$= -\frac{(a^2+b^2)}{a^2 e^2}$$

$$= -\frac{(a^2 + a^2(e^2-1))}{a^2 e^2} \quad \text{since } b^2 = a^2(e^2-1)$$

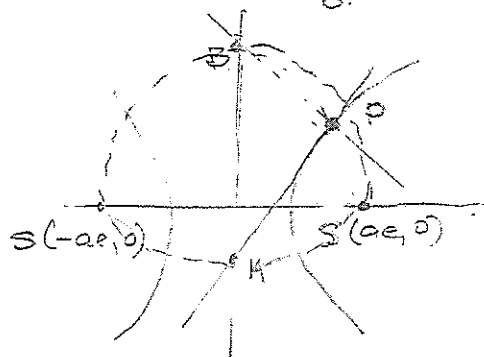
$$= -\frac{a^2 e^2}{a^2 e^2} = -1$$

$\therefore \angle BSA = 90^\circ$ and AB is the diameter of a circle passing through S (converse of $\angle$ in a semi-circle)

Similarly, $m_{AS'} \times m_{BS'} = \frac{-b}{ae \tan \theta} \times \frac{(a^2+b^2)\tan \theta}{abe}$

$$= -1$$

and the circle also passes through S'



QUESTION 16 (continued)

(b) (i) $1 - x^2 + x^4 - x^6 + \dots + (-1)^{n-1} x^{2n-2}$
is a G.P. $a=1$, $r=-x^2$ and there are "n" terms

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{1 \cdot (1 - (-x^2)^n)}{1 - (-x^2)} \\ = \frac{1 - (-1)^n x^{2n}}{1 + x^2}$$

$$(ii) \quad \frac{1}{1+x^2} - (1 - x^2 + x^4 - \dots + (-1)^{n-1} x^{2n-2}) \\ = \frac{(-1)^n x^{2n}}{1+x^2}$$

If n is even, $RHS = \frac{x^{2n}}{1+x^2} \leq x^{2n}$ since $1+x^2 \geq 1$

If n is odd, $RHS = \frac{-x^{2n}}{1+x^2} \geq -x^{2n}$ since $1+x^2 \geq 1$

$$\therefore -x^{2n} \leq \frac{1}{1+x^2} - (1 - x^2 + x^4 - \dots + (-1)^{n-1} x^{2n-2}) \leq x^{2n}$$

For all n

$$(iii) \quad \int_0^1 -x^{2n} dx \leq \int_0^1 \frac{1}{1+x^2} - (1 - x^2 + x^4 - \dots + (-1)^{n-1} x^{2n-2}) dx \leq \int_0^1 x^{2n} dx$$

$$\left[-\frac{x^{2n+1}}{2n+1} \right]_0^1 \leq \left[\tan^{-1} x \right]_0^1 - \left[x - \frac{x^3}{3} + \frac{x^5}{5} - \dots + (-1)^{n-1} \frac{x^{2n-1}}{2n-1} \right]_0^1 \leq \left[\frac{x^{2n+1}}{2n+1} \right]_0^1 \\ -\frac{1}{2n+1} \leq (\tan^{-1} 1 - \tan^{-1} 0) - \left[1 - \frac{1}{3} + \frac{1}{5} - \dots + (-1)^{n-1} \frac{1}{2n-1} \right] \leq \frac{1}{2n+1}$$

$$\therefore -\frac{1}{2n+1} \leq \frac{\pi}{4} - \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots + (-1)^{n-1} \frac{1}{2n-1} \right] \leq \frac{1}{2n+1}$$

(iv) As $n \rightarrow \infty$, $LHS \rightarrow 0$, $RHS \rightarrow 0$ $\therefore$ in the limit,

$$0 = \frac{\pi}{4} - \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots \right] = 0$$

$$\text{and } \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \dots$$